

Exercise 3

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Import all necessary packages needed to run the code

```
library(MASS)
library(ggplot2)
library(dplyr)
library(lme4)
data(stormer)
set.seed(42)
```

a)

We apply nonlinear regression to the given formula: first we solve the linear formula, then use the resulting parameters as an initial guess to the nonlinear regression. Since the linear model is already very close, nonlinear regression only needs 2 iterations.

```
##
## Call:
## lm(formula = form, data = stormer)
##
## Coefficients:
## (Intercept)      Viscosity          Time
##      220.138         28.099          2.082

t_lin = lin_model$coefficients
initial = c(theta1 = unname(t_lin[2]), theta2=unname(t_lin[3]))
form=as.formula(Time~ (theta1*Viscosity)/(Wt - theta2))
nmodel = nls(form, data=stormer,start=initial)
summary(nmodel)

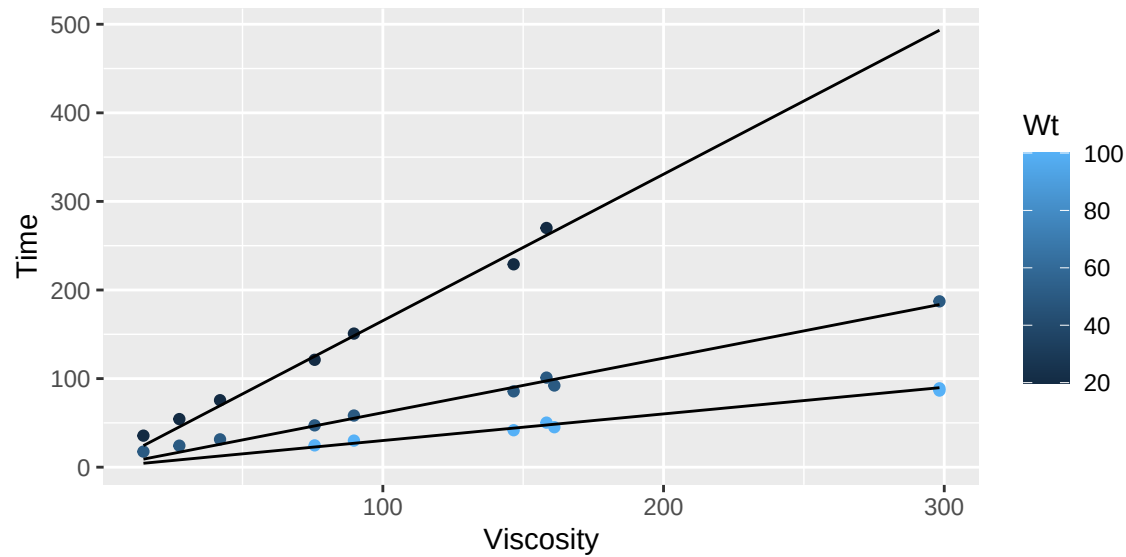
##
## Formula: Time ~ (theta1 * Viscosity)/(Wt - theta2)
##
## Parameters:
##      Estimate Std. Error t value Pr(>|t|)
## theta1  29.4013     0.9155  32.114 < 2e-16 ***
## theta2   2.2182     0.6655   3.333  0.00316 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.268 on 21 degrees of freedom
##
## Number of iterations to convergence: 2
## Achieved convergence tolerance: 9.828e-06
```

```
confint(nmodel)
```

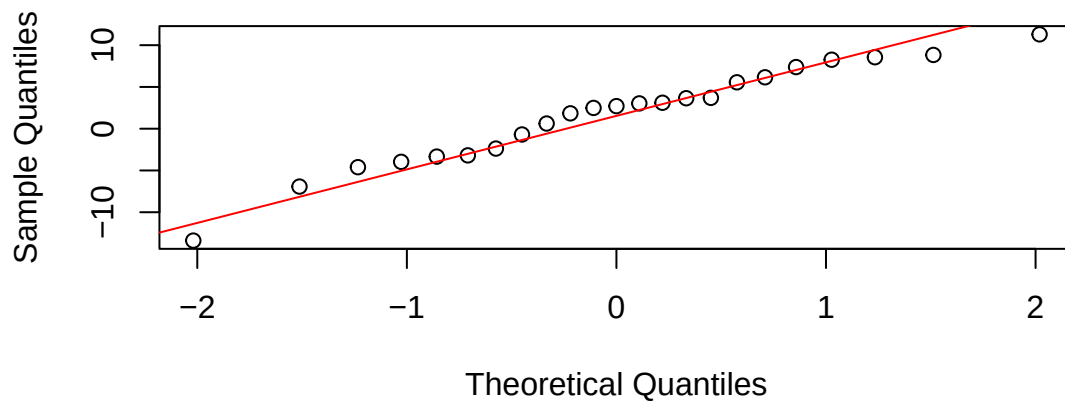
```
##           2.5%      97.5%  
## theta1 27.5439896 31.348359  
## theta2  0.7774809  3.545203
```

```
deviance(nmodel)
```

```
## [1] 825.0514
```



Normal Q-Q Plot



```
## sigma_squared  
##      39.28816
```

From visual inspection, the model fit looks accurate. The residuals also appear to follow an normal distribution as suggested by the qq-plot. The variance of e is computed above, and shows that there is significant measurement error in the data.

c)

Assuming Asymptotic normality, we can compute the confidence intervals for theta using the student distribution with N-2 degrees of freedom and the square root of the diagonals of the covariance matrix:

```
cov.est=vcov(nmodel);cov.est

##           theta1      theta2
## theta1  0.8382035 -0.5605465
## theta2 -0.5605465  0.4429206

coef(nmodel)

##      theta1      theta2
## 29.401298  2.218245
alpha = 0.08
for (i in 1:2){
  ci = coef(nmodel)[i]-qt(c(alpha/2,1-alpha/2),23-2)*sqrt(cov.est[i,i])
  print(ci)
}

## [1] 31.08556 27.71703
## [1] 3.4425747 0.9939153
```

d)

Similarly, we can get the confidence intervals for Time(Viscosity) at a constant Wt. For this we compute the gradient of our function at each point and assume a linear response.

```
alpha=0.006
grad50 = function(v, theta){
  rbind(theta[1]/(50 - theta[2]),
    (-theta[1]*v)/((50-theta[2])*(50-theta[2]))
  )
}
v3 = function(v) grad50(v, coef(nmodel))
cov.est

##           theta1      theta2
## theta1  0.8382035 -0.5605465
## theta2 -0.5605465  0.4429206

se = function(v) sqrt(t(v3(v)) %*% cov.est %*% v3(v))
ci = function(v) f_fit50(v) + (qt(c(alpha/2, 1-alpha/2),23) * (se(v)))
ub = function(v) ci(v)[1]
lb = function(v) ci(v)[2]

test_v = seq(from=10, to=300, by=5)
ub_data = numeric(length(test_v))
lb_data = numeric(length(test_v))
fv_data = numeric(length(test_v))
for (i in 1:length(test_v)){
  ub_data[i] = ub(test_v[i])
  lb_data[i] = lb(test_v[i])
  fv_data[i] = f_fit50(test_v[i])
}
```

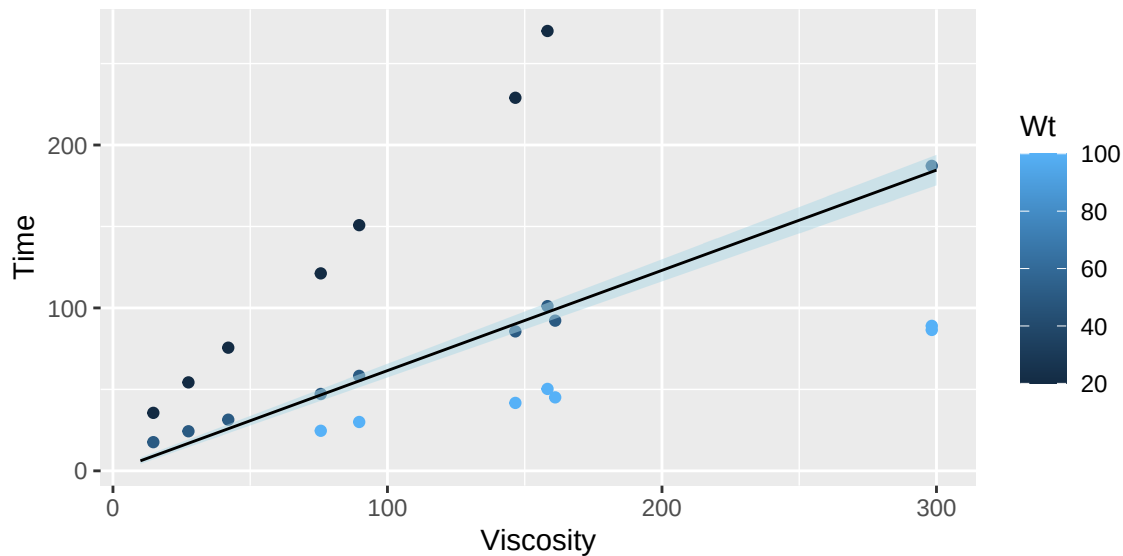
```

plot_data = data.frame(
  v = test_v,
  ub = ub_data,
  lb = lb_data,
  ub1 = ub_data,
  lb1 = lb_data,
  fv = fv_data
)

plot = ggplot() +
  geom_point(data=stormal, aes(x=Viscosity, y=Time, color=Wt)) +
  geom_ribbon(data=plot_data, aes(x=test_v, ymin=lb, ymax=ub), fill = "lightblue",alpha=0.5)+
  geom_line(data=plot_data, aes(x=test_v,y=fv))

plot

```

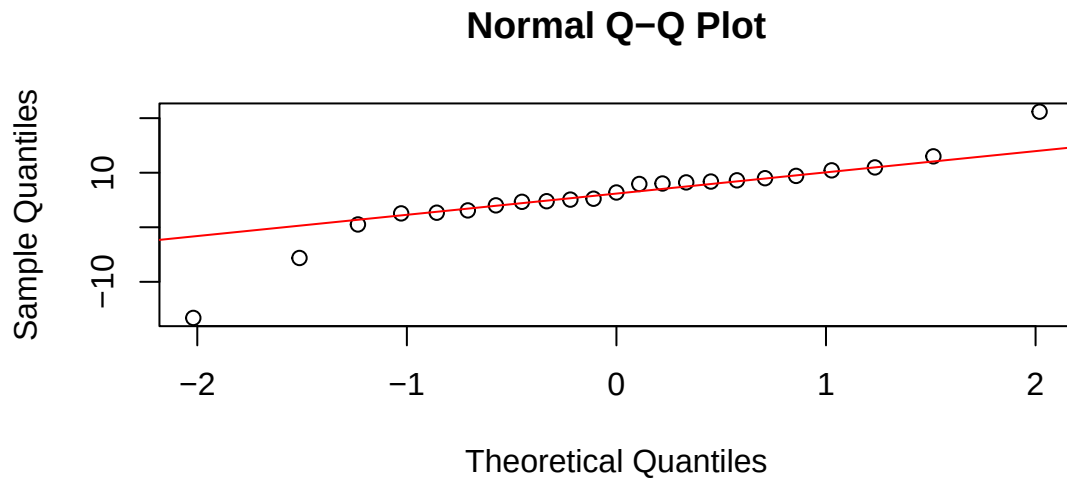


b), e)

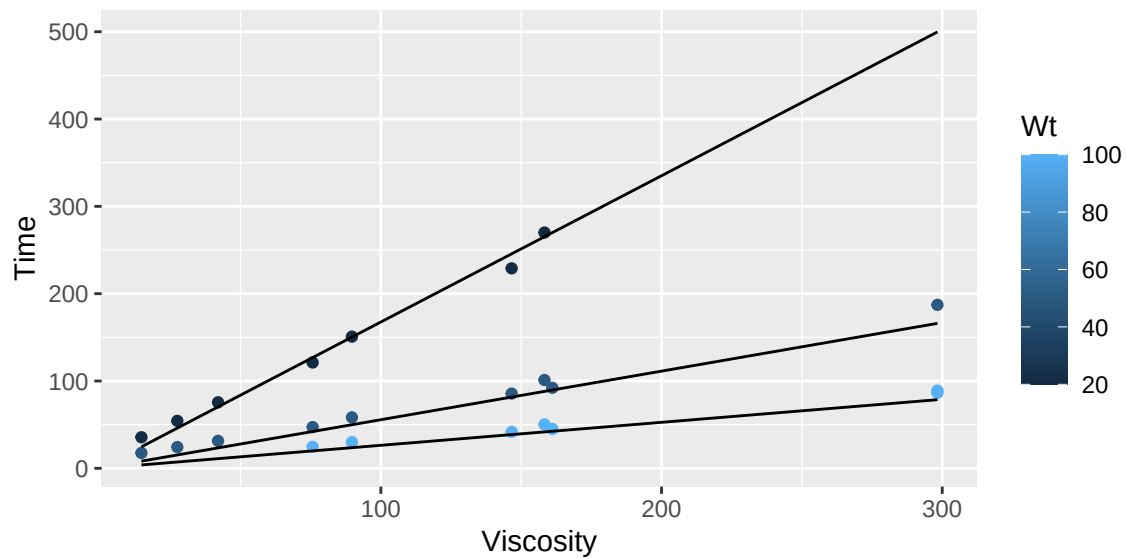
```

##
## Formula: Time ~ (25 * Viscosity)/(Wt - theta2)
##
## Parameters:
##      Estimate Std. Error t value Pr(>|t|)
## theta2    5.0786    0.3187   15.94 1.44e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.15 on 22 degrees of freedom
##
## Number of iterations to convergence: 4
## Achieved convergence tolerance: 4.23e-06

```



[1] 87.711



25 lies outside of our confidence interval for θ_1 . The limited model where $\theta_1 = 25$ is fixed shows a much higher residual sum of squares. The model is now also underestimating Time for Weights of (50, 100), only accurately representing the datapoints from $Wt=20$. Additionally, the Q-Q plot is now skewed slightly. Therefore we conclude that the $\theta_1=25$ model does not represent the data well.